

AL AQIQ, Saudi Arabia

WMO: 410550

Lat: 20.296N

Lon: 41.634E

Elev: 5486

StdP: 12.01

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.0	48.2	15.4	14.7	64.6	19.3	17.8	68.3	18.7	66.7	16.2	67.2	4.2	120	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	97.0	62.3	96.1	62.2	95.0	62.1	70.2	84.0	68.7	83.5	67.4	83.1	11.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	118.4	79.3	64.3	110.9	76.8	62.6	104.5	75.1	38.0	84.9	36.6	83.2	35.4	82.9	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	18.2	15.9	40.8	99.5	1.8	1.2	39.5	100.4	38.4	101.1	37.4	101.7	36.1	102.6
			33.4	72.5	1.4	1.2	32.4	73.4	31.6	74.1	30.8	74.8	29.8	75.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	74.5	62.2	65.9	69.8	74.0	79.7	84.6	85.1	82.1	74.2	67.5	62.8	(6)	
(7)		DBStd	9.05	4.61	4.50	4.11	4.00	3.58	2.09	2.42	1.88	2.46	2.77	3.31	3.96	(7)
(8)		HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	260	106	40	10	3	0	0	0	0	0	14	87		(9)
(10)		CDD50	8928	381	446	615	721	919	1037	1087	1088	962	750	524	398	(10)
(11)		CDD65	3712	21	65	160	274	454	587	622	623	512	286	88	20	(11)
(12)		CDH74	42016	155	455	1097	2138	4685	7715	8259	8186	6164	2431	585	146	(12)
(13)		CDH80	19871	11	55	213	696	2138	4128	4495	4412	2993	671	55	4	(13)
(14)	Wind	WSAvg	7.1	5.6	6.2	6.6	6.2	6.3	8.8	10.6	9.8	7.7	6.4	5.3	5.2	(14)
(15)	Precipitation	PrecAvg	5.2	0.5	0.0	0.4	1.5	0.9	0.2	0.4	0.4	0.1	0.4	0.3	0.1	(15)
(16)		PrecMax	11.1	4.3	0.1	2.5	5.2	2.7	0.9	4.2	1.0	0.6	2.2	1.7	0.7	(16)
(17)		PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.4	0.9	0.0	0.6	1.6	0.7	0.2	0.8	0.3	0.2	0.6	0.4	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.0	84.3	87.4	90.6	95.1	97.8	98.9	98.4	96.3	88.3	83.8	80.5	(19)
(20)			MCWB	57.0	58.1	60.4	60.1	61.5	62.2	63.2	62.9	62.8	58.5	59.6	56.5	(20)
(21)		2%	DB	77.8	82.0	84.4	88.1	93.3	96.6	97.1	96.8	94.2	86.7	81.8	77.4	(21)
(22)			MCWB	56.9	58.7	60.1	60.9	61.0	61.7	62.5	62.3	61.8	57.6	58.6	57.2	(22)
(23)		5%	DB	75.2	79.1	82.4	86.3	91.6	95.1	96.4	95.3	92.9	85.6	80.2	75.3	(23)
(24)			MCWB	57.1	58.3	59.6	61.8	61.2	61.4	62.4	62.6	60.9	57.3	58.7	58.0	(24)
(25)		10%	DB	72.7	76.9	80.4	84.3	89.8	93.6	94.7	94.6	91.4	84.0	77.4	73.2	(25)
(26)			MCWB	56.6	58.3	60.3	61.7	61.5	60.9	62.3	62.5	60.4	57.1	58.2	57.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	65.2	67.4	69.2	69.4	70.0	72.1	72.3	71.6	66.6	65.7	65.1	(27)
(28)			MCDWB	73.4	75.7	79.9	81.8	83.5	85.9	87.1	85.0	86.0	75.6	75.2	73.8	(28)
(29)		2%	WB	61.1	63.0	65.5	67.4	67.6	67.5	69.6	70.4	67.2	65.0	64.2	62.8	(29)
(30)			MCDWB	71.4	74.3	77.3	79.3	81.7	86.5	84.9	83.2	83.0	75.7	74.8	71.2	(30)
(31)		5%	WB	59.9	61.8	64.3	65.9	66.2	65.9	67.9	68.7	65.1	63.3	62.7	61.6	(31)
(32)			MCDWB	70.0	73.0	76.0	77.8	80.5	86.2	85.1	83.3	84.0	75.2	73.2	70.0	(32)
(33)		10%	WB	58.9	60.3	62.9	64.7	65.0	64.5	66.1	67.1	63.5	61.4	61.5	60.1	(33)
(34)			MCDWB	68.8	71.7	74.7	76.9	80.4	86.2	85.7	84.1	85.1	75.7	72.0	68.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.0	21.7	20.9	20.4	20.7	20.5	20.1	20.2	19.5	20.2	20.9	21.3	(35)
(36)			MCDBR	23.4	23.5	22.3	21.8	21.3	21.1	21.0	20.8	20.3	20.5	22.5	22.8	(36)
(37)			MCWBR	12.2	12.4	11.1	9.8	9.1	9.9	9.8	9.2	9.2	9.1	10.7	11.9	(37)
(38)		5% WB	MCDBR	20.5	21.4	19.4	20.2	20.6	20.1	20.2	19.9	19.3	19.9	20.3	19.8	(38)
(39)			MCWBR	12.2	12.8	10.3	9.5	9.2	10.9	10.9	9.9	10.0	10.7	10.6	11.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.370	0.408	0.454	0.514	0.546	0.579	0.593	0.512	0.404	0.373	0.344	(40)	
(41)		taud	2.390	2.273	2.146	2.029	1.851	1.755	1.717	1.684	1.833	2.134	2.267	2.372	(41)	
(42)		Ebn at Noon	296	293	285	272	253	243	235	233	251	278	282	291	(42)	
(43)		Edh at Noon	35	42	49	56	67	73	75	78	67	47	39	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1518	1787	2046	2165	2305	2387	2226	2099	2060	1958	1627	1468	(44)	
(45)		RadStd	100	95	91	114	107	68	78	58	62	64	69	98	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	-63	+277	+229	

Nomenclature: See separate page